CURRICULUM VITAE

Md. Humayun Kabir

Address

Place of Birth: *Cox's Bazar, Bangladesh*
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1. Key strengths and professional skills

My research interests include landscape ecology, urbanization, climate change, GIS, remote sensing, and GeoAI. My methodology combines science and social science tools/methods (e.g., GIS, satellite remote sensing, GPS techniques, computer coding, statistical methods as quantitative techniques, common property resource management and questionnaire survey, PRA, FGD as qualitative tools) to study land use change and climate impacts situation. Wide-ranging field experiences, national level contributions, experience to work with government agencies, donors and NGOs (I completed more than fifteen research projects for different agencies) are the pillars upon which my professional strength, integrity, and skills are relied. The following table gives impression about my research experiences with different agencies.

Name of the Organization	Study/work Title	Type of work	Role
Ministry of Science and Technology of Bangladesh (Ongoing)	1. Multivariate seasonal analysis of oceanographic parameters in four oceanic regions using satellite data and machine learning	This study will develop a comprehensive data-driven understanding of seasonal ocean dynamics in the Bay of Bengal and adjacent seas by leveraging satellite observations and modern ML methods	Associate Investigator
UN Women (Ongoing)	2. EmPower: Women for Climate Resilient Societies (Phase-II)	The objective of this study is to develop a gender-sensitive climate action plan	Climate Expert
Palli Karma-Sahayak Foundation (PKSF) (Ongoing)	3. Promoting Resilient Green Growth in Dairy Sub-sector through Ensuring RECP	Baseline Survey	PI
Palli Karma-Sahayak Foundation (PKSF) (Ongoing)	4. Adopting Technologies and Practices for Resilient Green Growth in Fisheries	Baseline Survey	PI
Palli Karma-Sahayak Foundation (PKSF)	5. Enhancing Income of Entrepreneurs through Production and Marketing of Mustard Safe Edible Oil Value Chain	Endline Survey	PI
World Health Organization	6. Vulnerability and adaptation assessment of climate change impacts on nutrition in Bangladesh	The study explores the impacts of climate change on nutrition in twelve climate stress areas of Bangladesh	Climatologist /Geographer
Winrock International	7. B-PEMS AugroJatra Climate Change Project	Developing training module on climate change and human trafficking, and facilitating training workshop for national and district level government officials	Climate Expert
Tara Climate Foundation	8. Energy Production and Air Pollution: Reviewing the	This study involved reviewing gaps of the Air Pollution	Research Team

	Emission Standards of Bangladesh	Control Rules 2022 of Bangladesh	Member
Department of Environment, Ministry of Environment, Forest and Climate Change, Government of Bangladesh	9. Bangladesh National Air Quality Management Plan 2024-2030	The objective of this task is to develop a national plan to reduce air pollution in Bangladesh which will be implemented between 2024 and 2030	Consultant
US Forest Service	10. Leaf Dust Accumulation in Plant Species in Dhaka City	The study analyzed the presence of lead rate in the road-side leaf dust in Dhaka city as an effort to formulate policies for lead contamination-free Bangladesh	Research Team Member
World Bank	11. Support to Cost-benefit Analysis of Interventions to Address Lead Contamination in Bangladesh	The study investigated the status, causes, sources, and consequences of lead contamination in Bangladesh to support policy formulation for lead contamination-free Bangladesh	Research Team Member
UNDP	12. Lessons Learned for Adaptive Livelihoods in South-western Waterlogging Areas of Bangladesh	The study critically examined the waterlogging problems and learned lessons from both success and failures of interventions implemented by different agencies	Co-PI
	13. Multi-Sectoral Needs Assessment for Recovery of Cyclone Roanu (2016) Impacts in Banskhali Upazila of Chittagong	The study examined the aftermaths of cyclone Roanu and provides an assessment of damage and loss in financial terms, also provided impact-recovery plan	Co-PI
Palli Karma-Sahayak Foundation (PKSF)	14. Development and Extension of Livestock (cow based) Cluster in Tala Upazila (Satkhira district) by Using Modern Technology and Adopting Improved Management Practice	Baseline survey	PI
Nabarun Publication	15. Ogni Geographic Atlas (English and Bangla version)	Atlas of Bangladesh	Cartographer
Asiatic Society of Bangladesh	16. National Atlas of Bangladesh	GIS and Remote Sensing database management	Cartographer
CEGIS	17. Land use zoning of Bangladesh	GIS and remote sensing data acquisition, data manipulation and validation, and map preparation	Junior Researcher

2. Educational Qualifications

Degree	Grade	Passing year	Department &Name of institution
PhD	Ongoing		Interdisciplinary Humanities, University of California, Merced, USA
M.Sc.	3.70/4	2015	Geography and Environment, Jahangirnagar University
B.Sc.	3.57/4	2014	Geography and Environment, Jahangirnagar University

HSC	5.00/5	2008	Govt. Hazi Muhammad Mohsin College
SSC	5.00/5	2006	Shilkhali High School

3. Thesis and Dissertation

Under-Graduate Dissertation: Cross Border Informal Trading in Teknaf Upazila: Pattern and Conditions.

M.Sc. Thesis: Application of GIS & Remote Sensing to Analyze the Coastal Erosion & Accretion at South-Eastern Coast of Bangladesh.

4. Career Summary

My major career attachments are,

Assistant Professor (February 2024 – present), Department of Environmental Science, Stamford University Bangladesh.

Assistant Director (April 2024 – present), Center for Atmospheric Pollution Studies (CAPS).

Lecturer (January 2017 – January 2024), Department of Environmental Science, Stamford University Bangladesh.

Part-time Faculty (August 2017 – February 2019), Institute of Remote Sensing and GIS, Jahangirnagar University, Bangladesh.

Junior Researcher (February 2015 – December 2016), Centre for Environmental and Geographic Information Services (CEGIS), Bangladesh.

5. Selected Academic Publications

Rashid, M., **Kabir, M.H.** and Anonna, M. 2026. The Trend of Agricultural Yields in the Context of Climate Change in Bangladesh, *Field Crops Research* (Submitted).

Kabir, M.H., Jahan, N., Amin, A., Sultana, S. and Oishi, I.J. 2026. Historical Channel Dynamics of the Teesta River: A Geospatial Investigation, *Nature Communication* (Submitted).

Kabir, M.H. and Ryavec, K.E. 2026. Historical GIS of British India in 1891: Society, Economy, and Urbanization, *Journal of the Royal Asiatic Society*, ISSN 1356-1863 (Print)/1474-0591 (Online) (Submitted).

Khatun, K., Hossain, M.S., **Kabir, M.H.** and Majumder, A.K. 2019. Climate Change in Bangladesh: a closer look to temperature and rainfall data, *Climate Change Journal*, Vol. 5(20), ISSN 2394-8558.

Latifa, G.A., Majumder, A. K., **Kabir, M.H.** and Chakma, A. 2019. Water Quality and Fish Diversity of Chengi River of Khagrachari District, *Bangladesh Journal of Zoology,* Vol. 47(2), ISSN 0304-9027 (Print)/ 2408-8455 (Online).

Kabir, M.H., Rahaman, H. and Afroze, M. 2018. Vegetation Cover Change Detection and Its Impact on Human Health in Dhaka City, Bangladesh, *Stamford Journal of Environment and Human Habitat*, Vol. 7, ISSN 2226-8898.

Kabir, M.H. 2018. Flood (2017) in North-Eastern Haor Region of Bangladesh: Causes and Consequences, *Journal of Disaster Management,* Begum Rokeya University Rangpur, Vol. 2.

Kabir, M.H., Alam, S.M.S. and Elahi, K.M. 2017. Application of GIS for Multi-hazard Mapping of Dakshin Khali Union, Bagerhat, *Stamford Journal of Environment and Human Habitat*, Vol. 6, ISSN 2226-8898.

Kabir, M.H., Sobnam, M. and Islam, M.N. 2016. Using Remote Sensing and GIS to Detect Green Space Dynamics at Dhaka City, Bangladesh, The *Jahangirnagar Review,* ii: Social Sciences, xxxvii, Jahangirnagar University, Bangladesh, ISSN-1682-422.

Kabir, M.H., Sobnam, M. and Islam, M.N. 2016. River Flow Regime in Coastal Erosion and Accretion at South-Eastern Coast of Bangladesh, *Bangladesh National Geographical Association Journal*, Vol. 39 & 40.

Kabir, M.H., Sobnam, M. and Hossain, M.A. 2016. Analysis of Insect Pests Carried by Imported Wheat in Bangladesh, *Stamford Journal of Environment and Human Habitat*, Vol. 5, ISSN 2226-8898.

Sobnam, M., **Kabir, M.H.** and Islam, M.N. 2016. Spatial and Temporal Variation of Nor'wester in Bangladesh: A Geographical Analysis, *Somaj Biggan Samikhya*, Vol. 2, Faculty of Social Science, Jahangirnagar University, ISSN 2309-6934.

Sobnam, M., **Kabir, M.H.** and Islam, M.N. 2015. Seasonal Variability of the Teesta River in Bangladesh, *Stamford Journal of Environment and Human Habitat*, Vol. 4, ISSN 2226-8898.

Kabir, M.H., Sobnam, M. and Islam, M.N. 2015. Cyclonic Disturbances Due to Climate Change. *Jahangirnagar Bishwavidyalay Bhugol O Paribesh Samikkhan*, 33, Department of Geography and Environment, Jahangirnagar University, Bangladesh, ISSN-10278567.

Kabir, M.H., Sobnam, M. and Islam, M.N. 2015. Application of GIS and Remote Sensing to Analyze the Coastal Erosion and Accretion at South-Eastern Coast of Bangladesh, *The Jahangirnagar Review,* ii: Social Sciences, xxxvi, Jahangirnagar University, Bangladesh, ISSN-1682-422.

Kabir, M.H. 2014. Cross Border Informal Trading in Teknaf Upazila: Pattern and Conditions. *Jahangirnagar Bishwavidyalay Bhugol O Paribesh Samikkhan*, 32, 2013, Department of Geography and Environment, Jahangirnagar University, ISSN-10278567.

6. Invited Lectures, Conference Presentations and Publications in the Proceedings

I have presented at conferences and workshops to a range of different audiences including academic and the public. Some of the presentated papers are also published in the conference proceedings. Major presentations are highlighted here.

Presented the research paper titled "**Rohingya Refugee-Driven Land Use/Land Cover Change and Landslide Susceptibility in Cox's Bazar, Bangladesh: An Integrated Geospatial and Machine Learning Analysis"** at the Young Researchers Conference on Disaster Risk Reduction (DRR) 2026, held at Universiti Sains Malaysia, Penang, Malaysia, on May 06, 2026. The conference was jointly organized by Universiti Sains Malaysia, Ajman University, Chongqing University, iUNRC, DIGIHAZ, & GeogMalaysia.

Paper presented on "**Urbanization-Driven Land Use/Land Cover Change and Ecosystem Service Value Dynamics in Dhaka City (1991-2022): A Geospatial-Machine Learning Analysis"** at the 8th International Conference on Sustainable Development (ICSD 2026), organized by UIU and IDSS.

Paper presented on "**Modeling Urban Expansion and Ecosystem Service Value Loss in Dhaka City (1991-2050): A GeoAI-Based Remote Sensing Approach"** at the 8th International Conference on Sustainable Development (ICSD 2026), organized by UIU and IDSS.

Paper presented on "**Spatiotemporal Surface Water Dynamics in Brahmanbaria district, Bangladesh: A GIS-Remote Sensing Investigation"** at the 8th International Conference on Sustainable Development (ICSD 2026), organized by UIU and IDSS.

Paper presented on "**Impacts of Rohingya Refugee-Induced LULC Change on Surface Runoff Dynamics in Cox's Bazar, Bangladesh: An Integrated Machine Learning and Remote Sensing Approach"** at BAPA-BEN National Conference 2026 on Environmental Reforms in Bangladesh: Necessity and Task, organized by BAPA and BEN.

Paper presented on "**Agricultural Yield Trends in a Changing Climate: Evidence from Bangladesh"** at BAPA-BEN National Conference 2026 on Environmental Reforms in Bangladesh: Necessity and Task, organized by BAPA and BEN.

Delivered a public lecture on **"Cities from Space: Monitoring Urban Growth, Climate Change, and Urban Heat Island Intensity"** held on October 19, 2024 at the MASLab in Dhaka, organized by the Society for the Popularization of Science Bangladesh (SPSB).

Presented paper on **"Revolutions and Religion: A Core-Periphery GIS Model of the Historical Growth and Spread of Islam in Bangladesh circa 1204-1947"** at the Graduate Student Conference 2020 on Religions and Revolutions, University of California, Merced, USA, December 07, 2020.

Presented paper on **"Climate Change and Its Impact on Sundarbans"** at Bughol o Paribesh Conference 2019 on Climate Change and the Sundarbans, Kolkata, India, held from April 15-18, 2019.

Paper presented on "**Climate Change Adaptive Feeding System for Livestock Improvement in Satkhira District, Bangladesh"** at BGS Conference 2019 on Resource Management, Sustainability and Climate Change, organized by Bangladesh Geographical Society (BGS).

Paper on **"Change Detection of Water Bodies in Dhaka City: An Analysis Using GIS and Remote Sensing"** presented at BAPA-BEN International Conference 2019 on Delta Plan 2100 and Sustainable Development in Bangladesh, organized by BAPA and BEN.

Presented paper on **"Climate Variation (Temperature and Precipitation) In Bangladesh (1901-2015) And Prediction for Year 2050"** at BAPA-BEN International Conference 2019 on Delta Plan 2100 and Sustainable Development in Bangladesh, organized by BAPA and BEN.

Paper presented on "**Prospects and Constraints of Developing Tourism Industry at Sajek Valley, Rangamati"** at BAPA-BEN International Conference 2019 on Delta Plan 2100 and Sustainable Development in Bangladesh, organized by BAPA and BEN.

Paper presented on "**Assessing the Causes and Consequences of Landslide in Chittagong Hill Districts of Bangladesh"** at BAPA-BEN International Conference 2019 on Delta Plan 2100 and Sustainable Development in Bangladesh, organized by BAPA and BEN.

Presented paper on **"Investigation of Particulate Matter Concentration in the air of Khagrachari district, Bangladesh"** at 1st International Conference on Biological and Environmental Research: Recent Development, Challenges and Future Prospects, organized by Jessore University of Science and Technology, Jessore, Bangladesh.

Paper on **"Assessing the Water Quality and Fish Diversity of Chengi River at Khagrachari, Bangladesh"** presented at the 15th Bangladesh International Geographical Conference, organized by Bangladesh National Geographical Association (BNGA).

Presented research results on **"Application of GIS and Remote Sensing to Analyze the Coastal Erosion and Accretion at South-Eastern Coast of Bangladesh"** at 15th Bangladesh International Geographical Conference, organized by Bangladesh National Geographical Association (BNGA).

Paper presented in BAPA-BEN Special Conference on Sustainable Development Goals and Environment 2018 on "**Using Remote Sensing and GIS to Analyze River Bank Erosion at Ghorautra River, Kishoreganj,"** organized by BAPA and BEN.

Presented paper on **"Using Remote Sensing and GIS to Detect Green Space Dynamics at Dhaka City, Bangladesh"** at BAPA-BEN Special Conference on Sustainable Development Goals and Environment 2017, organized by BAPA and BEN.

7. International and National Training/Workshop/Seminar Attended

Attending in 6-week-long training course on **"GIS for Climate Action"** from February 18-April 01 2026, conducted by Environmental Systems Research Institute (ESRI), Inc., USA.

Attended in 6-week-long (12-hour) non-credit course on **"Oral Communication Skills for International Scholars"** in Fall 2019, conducted by English Language Institute, Center for Engaged Teaching and Learning (CETL), University of California, Merced, USA.

Attended in a 5-day-long *training of the trainer on* ***"Google Earth Engine"*** in June 2018, jointly organized by ICIMOD, NASA and USAID in ICIMOD, Nepal.

Attended in 5-day-long *"Training Workshop on* ***"Google Earth Engine"*** in August 2017, jointly organized by IRS-JU, ICIMOD, NASA and USAID in Jahangirnagar University, Bangladesh.

Attended in a 12-day-long Certificate Course on ***"Spatial Analysis Using R and Python,"*** organized by Institute of Remote Sensing and GIS, Jahangirnagar University, Bangladesh.

Attended in 6-day-long ***"Faculty Training Program,"*** organized by Institutional Quality Assurance Cell (IQAC), Stamford University Bangladesh.

Attended in a workshop on ***"Research Methodology,"*** organized by South Asian Youth Research Institute for Development (Sayrid) & South Asian Journal of Social Science (SAJSS) in Dhaka University, Bangladesh.

Attended in a workshop on ***"Self-Assessment Report (SAR) and Post SA Improvement Plan (PSAIP),"*** organized by Institutional Quality Assurance Cell (IQAC), Stamford University Bangladesh, Dhaka, Bangladesh.

Attended in a Workshop on ***"Assessment of Water Quality,"*** organized by Department of Environmental Science, Stamford University Bangladesh.

Attended in a 2-day Training on ***"ISO 14001:2015,"*** organized by Department of Environmental Science, Stamford University Bangladesh.

Attended in a Seminar on ***"Nature, Environment and Biodiversity Conservation and Development,"*** jointly organized by the Departmental of Environmental Science, Stamford University Bangladesh and Department of Environment, Ministry of Forest and Environment, Government of the People's Republic of Bangladesh.

Attended in 3-month-long non-credit **"Computer course"** in Fall 2010, conducted by the Department of Geography and Environment, Jahangirnagar University, Bangladesh.

8. Computer Skills

Operating System	Windows, Mac
Office Application	MS Word, MS Excel, MS PowerPoint and MS Picture Manager
Geographic Information System (GIS)	Arc GIS – 10x, Arc GIS Pro, QGIS, AutoCAD
Remote Sensing/ Image Analysis	ERDAS Imagine – 16x, Google Earth, Google Earth Engine (GEE), Open Street Map (OSM)
Statistical Software	SPSS, STATA
Programming Language	R, JavaScript, Python
Graphics Software	Adobe Photoshop, Adobe Illustrator

9. Language Proficiency

Mother tongue	Bengali

Second language English
(Engage freely in discussion, read & write more advanced materials)

10. Professional Memberships

Member, American Association of Geographers (AAG)
Life Member, National Oceanographic and Maritime Institute (NOAMI)
Member, Bangladesh National Geographical Association (BNGA)

11. Awards, Prizes and Special Events

- Awarded a research project as Associate Investigator worth $6000 from the Ministry of Science and Technology of Bangladesh for 2025-2026 cycle, titled "Multivariate seasonal analysis of oceanographic parameters in four oceanic regions using satellite data and machine learning"
- Full funded PhD Scholarship from the University of California, Merced, USA
- Awarded $1500 conference travel grants from the Interdisciplinary Computational Graduate Education Program (NRT ICGE) of the University of California, Merced.
- Awarded Spring 2021 Fellowship from the Interdisciplinary Humanities Graduate Group, University of California-Merced
- Awarded Fall 2020 Fellowship from the Interdisciplinary Humanities Graduate Group, University of California-Merced
- Awarded Summer 2020 Fellowship from the Interdisciplinary Humanities Graduate Group, University of California-Merced
- Received research award for *Ogni Geographic Atlas* from Nabarun Publication in 2014
- Awarded Jahangirnar University merit scholarship for undergraduate academic results
- Awarded Jahangirnar University merit scholarship for M.Sc. academic result

12. References

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